This document will contain the description of the extention activities for each task, as well as a solution for the nuclear problem sheet typed out in LaTeX.

Programming language and useful modules

For all 10 of our tasks, we used Python to code and simulate mainly beacuse it's readable and easy to use, as it's a high level language that is relatively mostly composed of familiar English words, and has a clear and logical syntax. It's also easy to use, which speeds up the process of coding. We used VScode as the environment since it's responsive and accessible, with a clear colour coding scheme.

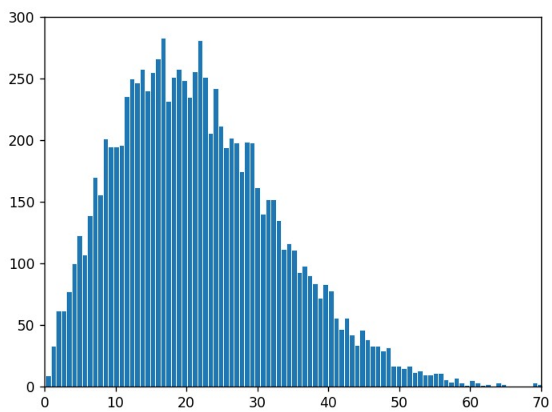
For almost all of the tasks, we used Matplotlib to create the animated and interactive simulations, and the main reason is its versatility. With it, we created not only graphs but also interactive coloured animations with sliders. We had to frequently work with arrays so Numpy was also extremely useful as the ndarray object provides many useful functions.

Task 1: Random walk

Create a model of a random walk of n steps of size s. Each step is in a random direction, with angle θ from the horizontal chosen from a uniform distribution between 0 and 2π radians.

```
for i in range(num_walks):
        angles = np.random.uniform(0, 2 * np.pi, N)
        x = np.concatenate([[0], np.cumsum(step_size * np.cos(angles))])
        y = np.concatenate([[0], np.cumsum(step_size * np.sin(angles))])
        l, = ax.plot(x, y, color=colors[i], linewidth=0.5, alpha=0.8)
        drawn.append(l)
```

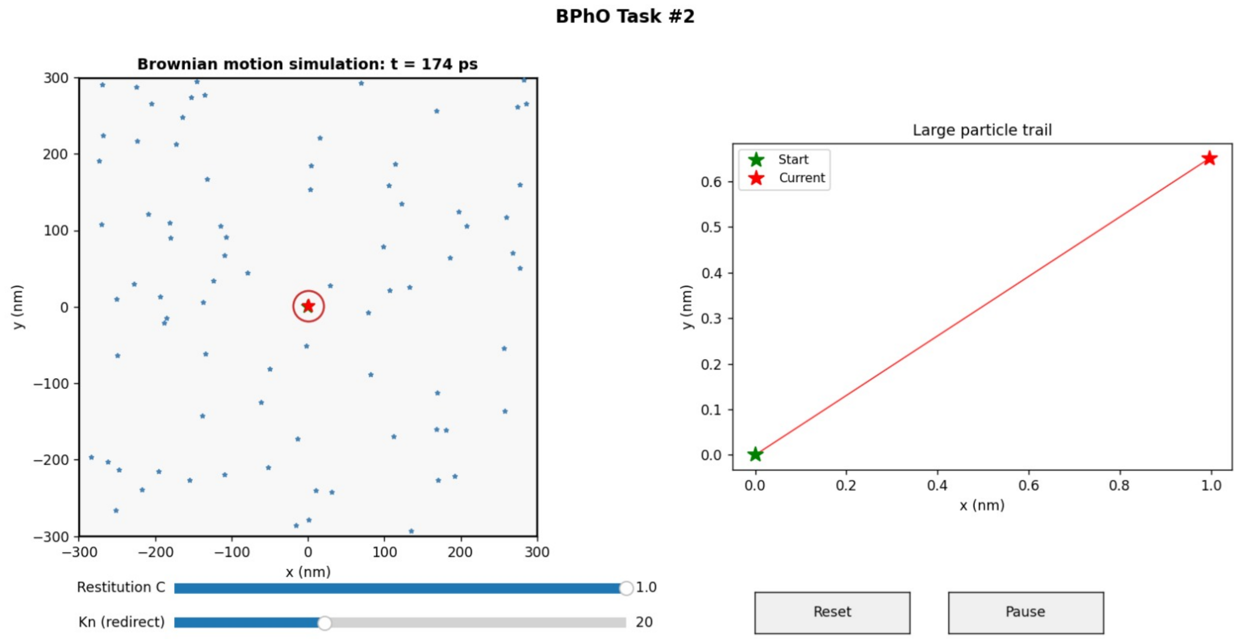
As shown in the code, for each step we generated a uniformly distributed random angle, then converted it into horizontal and vertical distances using $\Delta x = s \sin\theta$ and $\Delta y = s\cos\theta$.
We've also included a slider that controls how many walks are generated, and display them all at once. We can see that most walks end close to the origin, and a few stray walks drift far apart. Intrigued by this observation, we plotted a graph showing the distribution of displacement from the origin after the random walk.



The chart of number of walks out of 100000 against final displacement after 600 steps verifies the observation.

Task 2: Brownian motion

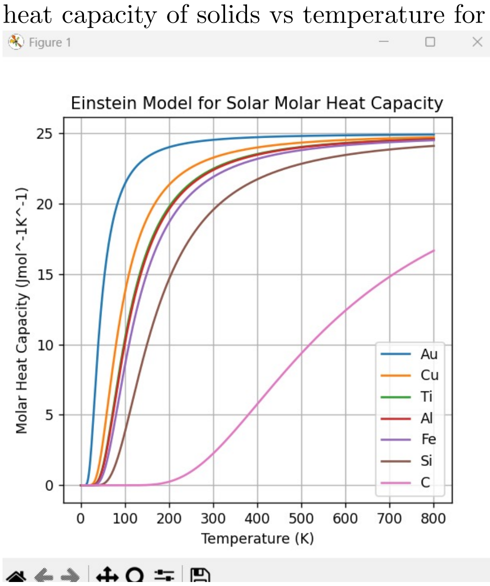
Consider N small particles of mass m and radius r moving randomly, some of which collide with a large particle of mass M and radius R. Use a random walk to model the smaller particles, and conservation of momentum to model collisions with the larger particle. Animate the subsequent motion of the system.



On top of finishing the task described, we also added an extra parameter of the restitution coefficient, and a graph showing the trail of the large particle.

Task 3: Planck "Black Body radiation" spectrum

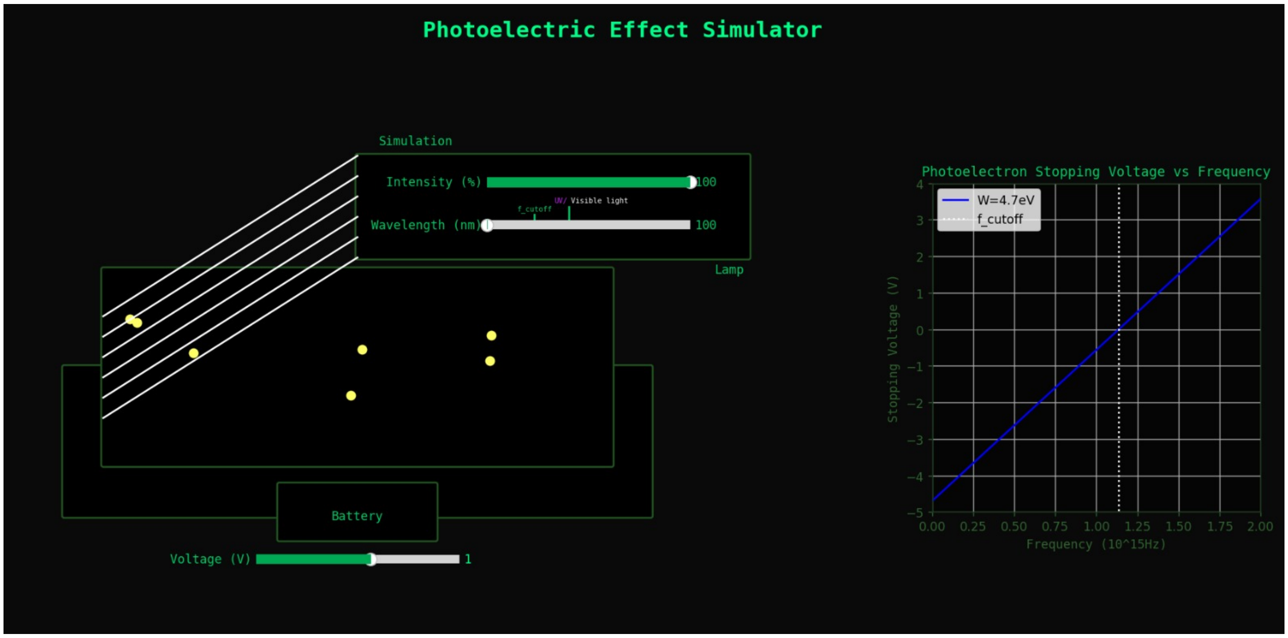
Plot the Planck "Black Body radiation" spectrum $B(\lambda, T)$ for several temperatures, and also plot Einstein's model of molar heat capacity of solids vs temperature for various atomic crystals such as gold, copper, iron.



For this task, we simply used matplotlib.pyplot module to plot the two required graphs.
We also colour coded the different lines in the same graph for readability as shown.

Task 4: Photoelectric effect

Plot photoelectron stopping voltage vs frequency (or in-vacuum wavelength) of incident photons for various metals.

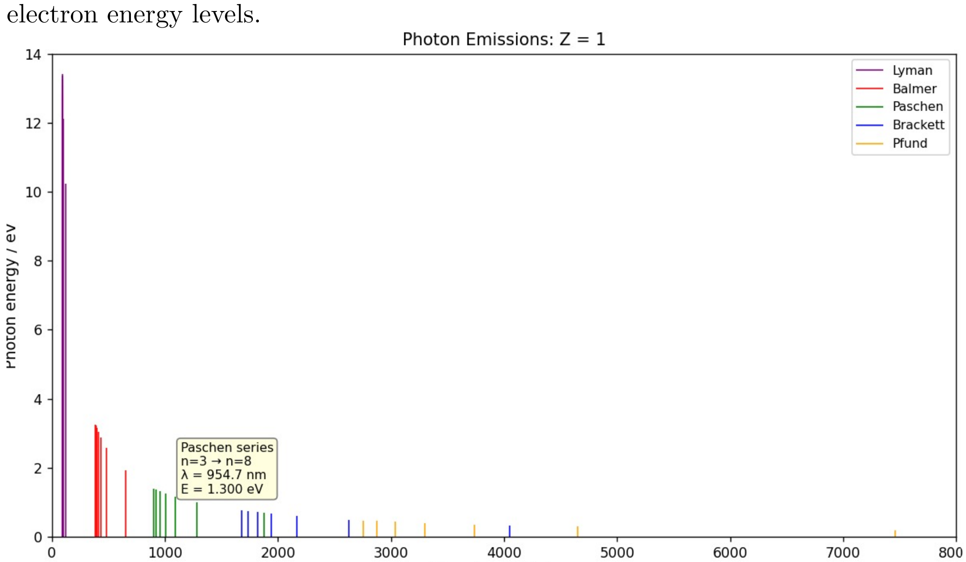


The graph is a simple blue straight line with a clear threshold frequency represented by the white dotted line. As an extension, we created an interactive simulation of the photoelectric effect, with parameters like intensity, wavelength and potential difference available to be changed.

The electrons are moddled as small yellow circles, and we can see that by increasing the intensity, the photoelectrons are emitted more frequently. We also moddled the speed of the electrons randomly with a maximum. Decreasing the wavelength will increase the maximum kinetic energy. Above threshold wavelength, there is no electron emission as not enough energy is provided.

Task 5: Hydrogen emission spectrum and Bohr's model of a hydrogenic atom

Create a graph of photon energy vs wavelength for photon emissions from hydrogen atoms due to transitions between electron energy levels.

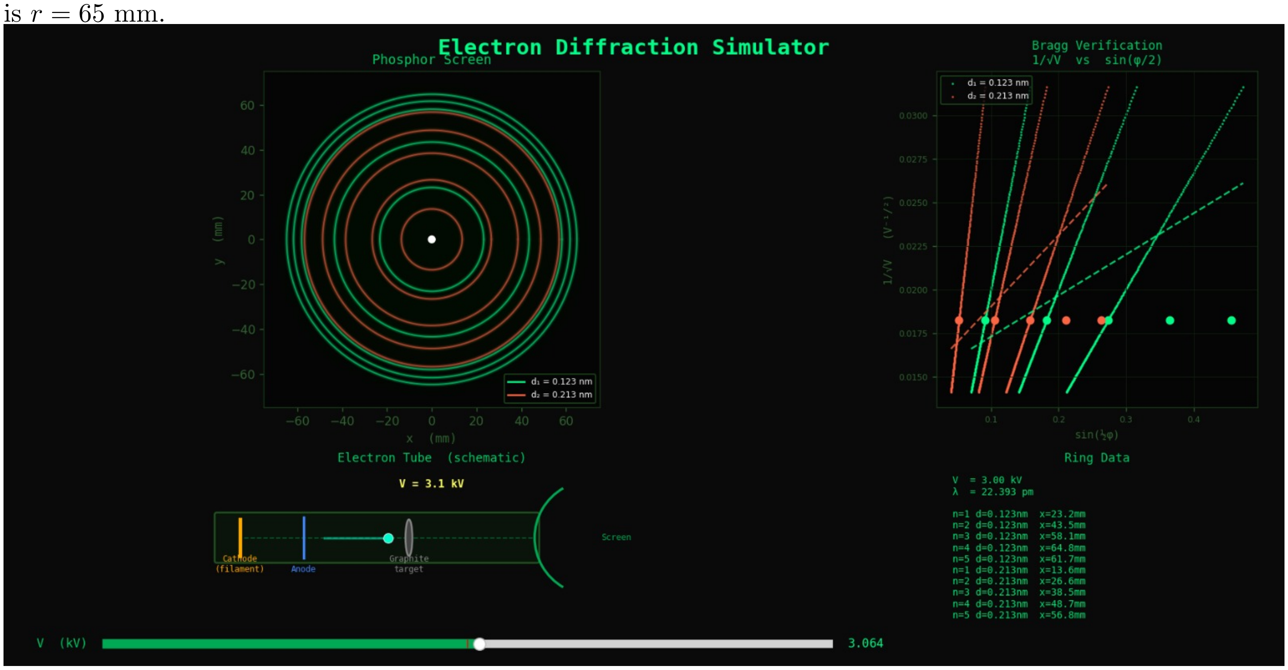


This task is again a simple graph. We added a special feature that shows the transition details when you hover over a line in the graph.

Task 6: Electron Diffraction

Create a computer model of the 'electron wave' rings on a phosphor screen with accelerating voltage V as a variable.

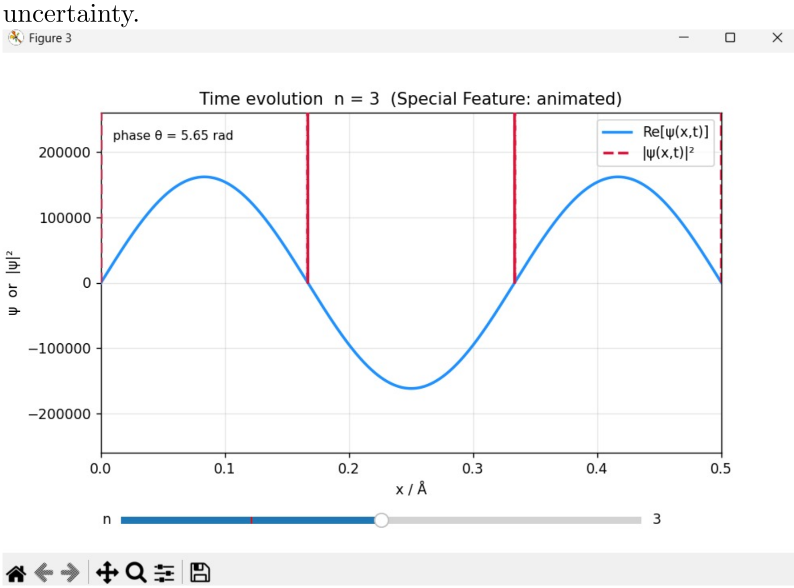
(1kV to 5kV). Assume atomic spacings d are layer spacings of graphite e.g. 0.123 nm or 0.213 nm. Check your model by generating a $\frac{1}{\sqrt{V}}$ vs $\sin\frac{1}{2}\phi$ straight line graph, where ring radius is $x = r\sin 2\phi$ and spherical glass diffraction tube radius is $r = 65$ mm.



The electron wave rings are shown on the left hand side, and the plot of $\frac{1}{\sqrt{V}}$ against $\sin\frac{1}{2}\phi$ is indeed a straight line for each of the rings, verifying the relationship $\sin\frac{1}{2}\phi = \frac{nh}{2d\sqrt{2meV}}$. On top of that, an electron is animated at the bottom left as it passes through the filament, anode and graphite target. A slider is added so that the voltage can be changed.

Task 7: Particle-in-a-box solution to the Schrödinger Equation

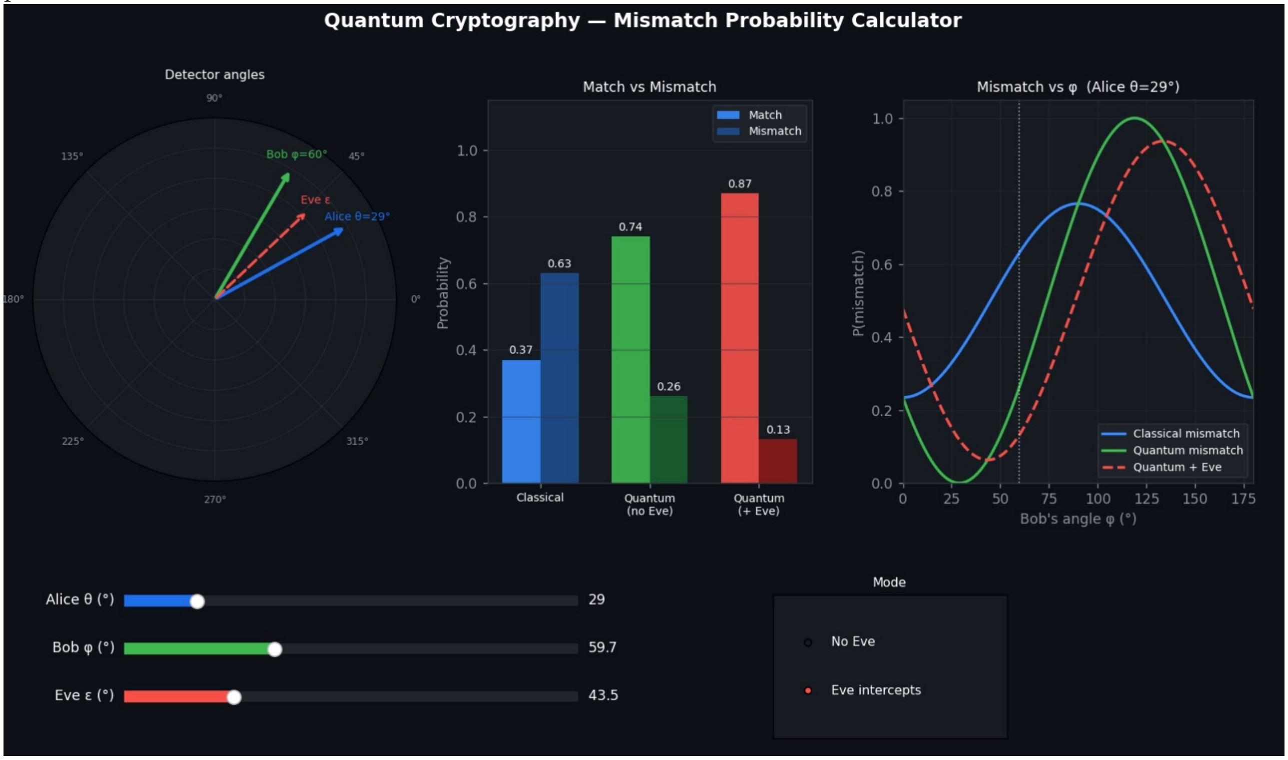
Plot energy vs quantum number, and probability densities $|\psi|^2$ vs displacement x in the box. EXTENSION: Write up the model, and show that a particle in the box obeys the Uncertainty Principle i.e. $\Delta x \Delta p \geq \frac{1}{2}\hbar$, where Δp is the momentum uncertainty.



On top of plotting the two straightforward graphs required, we additionally animated the wavefunction time evolution as graph of probability density against displacement, that is, the quantum particle state animated with respect to time. The figure shows the graph for $n = 3$.

Task 8: Quantum Cryptography

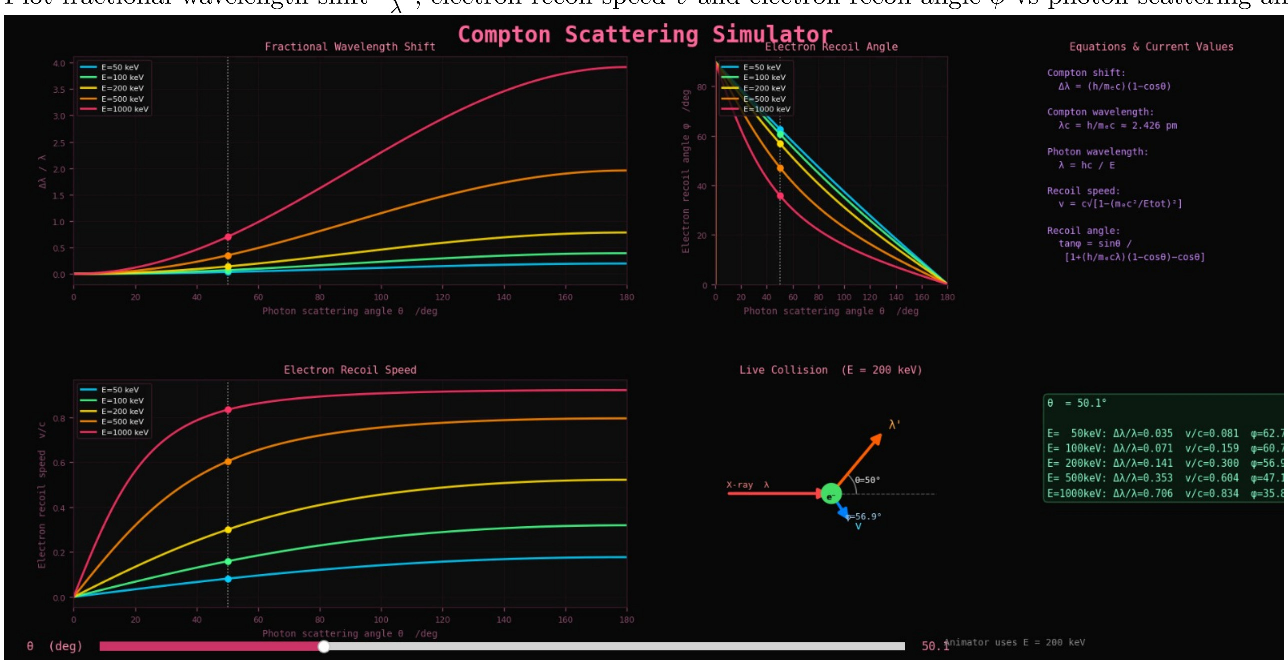
Create a visual calculator of the classical and quantum mismatch probabilities for the detection of polarized entangled photons.



As an extension task, we added a third detector Eve who measures at angle ϵ and showed how the mismatch probability changes with it. This is given by $P(\text{match}) = P_{AE} \cdot P_{EB} + (1 - P_{AE})(1 - P_{EB})$ where $P_{AE} = \cos(\epsilon - \theta)$ and $P_{EB} = \cos(\phi - \epsilon)$, changing the mismatch probability.

Task 9: Compton Scattering

Plot fractional wavelength shift $\frac{\Delta\lambda}{\lambda}$, electron recoil speed v and electron recoil angle ϕ vs photon scattering angle θ.



To plot the graph, simply use the formulae:

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

$$\lambda = \frac{hc}{E_\lambda}$$

$$v = c\sqrt{1 - (\frac{m_e c^2}{\frac{hc}{\lambda} - \frac{hc}{\lambda'} + m_e c^2})^2}$$

$$\tan\phi = \frac{\sin\theta}{1 + \frac{h}{m_e c\lambda}(1 - \cos\theta) - \cos\theta}$$

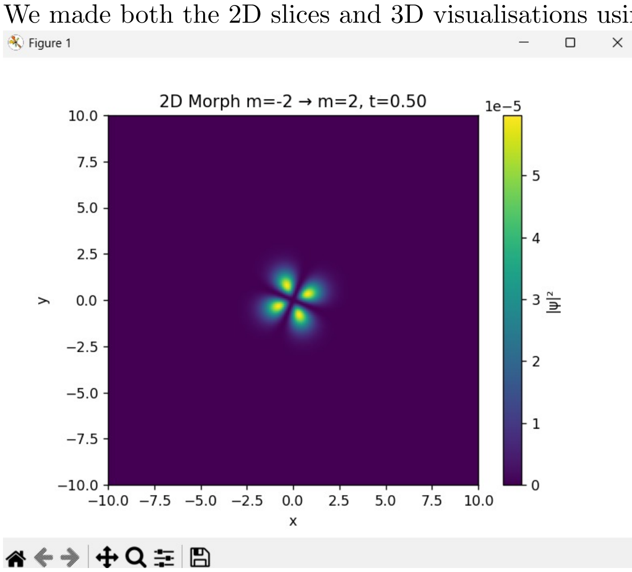
.

We also made a visualisation of the compton effect at $E = 200\text{keV}$ with matplotlib, with a slider that lets you vary the photon scattering angle θ, so that λ' and ϕ are changed accordingly. Finally the green chart on the right shows the values of the different expressions for different values of E.

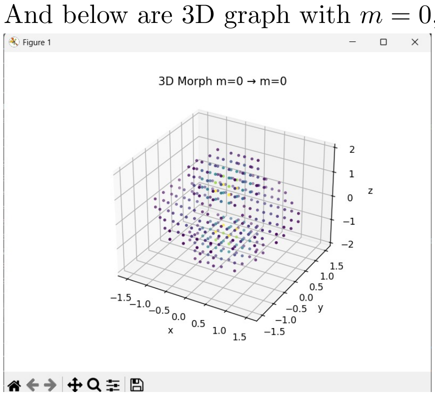
Task 10: Hydrogenic orbitals

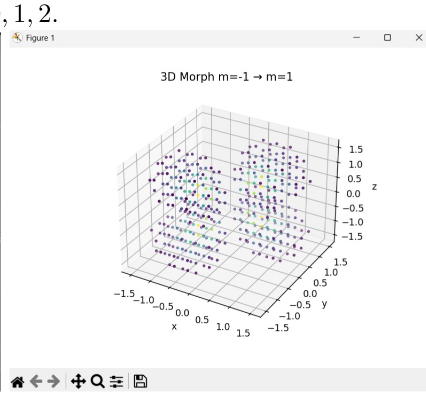
Plot 2D slices and 3D visualizations of the probability density $|\Psi|^2$ for an electron in a hydrogenic atom, given n, l, m quantum numbers.

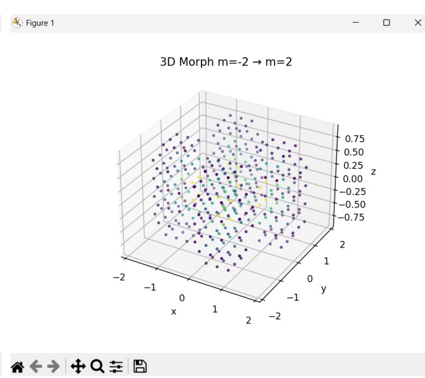
We made both the 2D slices and 3D visualisations using matplotlib. Below is a 2D slice:



And below are 3D graph with $m = 0, 1, 2$.







Nuclear Problems

1. (i) Substituting $A = 92, Z = 36, B = 764.77\text{MeV}$ into $M_x c^2 + B = Zm_p c^2 + (A-Z)m_n c^2$ gives

$$M_x + \frac{764.77 \times 10^6 \times 1.602 \times 10^{-19}}{(2.998 \times 10^8)^2} = 36 \times 1.6726 \times 10^{-27} + (92-36) \times 1.6749 \times 10^{-27} \quad (1)$$

$$\implies \quad M_x = 1.526 \times 10^{-25}\text{kg} = 91.927u \quad (2)$$

(ii) We know ${}^{235}_{92}\text{u} + {}^{1}_{0}\text{n} \to {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3 \times {}^{1}_{0}\text{n}$, and $\Delta E = B_{Ba} + B_{Kr} - B_u$, therefore we have

$$\begin{aligned} B_u &= B_{Ba} + B_{Kr} - \Delta E && (3)\\ &= (1145.36 + 764.77 - 173.28)\text{MeV} && (4)\\ &= 1736.85\text{MeV} && (5)\end{aligned}$$

(iii)(a) Energy released/kg of u-235 $= \frac{173.28 \times 10^6 \times 1.602 \times 10^{-19}}{235 \times 1.6605 \times 10^{-27}} = 7.11 \times 10^{13}\text{Jkg}^{-1}$
(b) (Energy density of u-235)/(Energy density of natural gas)$= \frac{7.11 \times 10^{13}}{4.9 \times 10^7} = 1.45 \times 10^6$
(c) Mass of u-235 $= \frac{10^{19}}{7.11 \times 10^{13}} \frac{1}{1000} = 140.6$tonnes
Density of uranium$= 19100\text{kgm}^{-3}$
Therefore, Volume$= \frac{140.6 \times 10^3}{19100} = 7.36\text{m}^3$
(iv) We have ${}^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \to {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$ and

$$\begin{aligned} B_D &= m_p c^2 + m_n c^2 - 2.0141\frac{u}{c^2} && (6)\\ &= (938.3 + 939.6 - 2.0141 \times 931.5)\text{MeV} && (7)\\ &= 1.7659\text{MeV} && (8)\end{aligned}$$

$\therefore B_T = B_{He} - B_D - \Delta E = (27.27 - 1.7659 - 17.59)\text{MeV} = 7.91\text{MeV}.$
(v) For $2u$ of deuterium and $3u$ of tritium, Deuterium-Tritium fusion reaction produces 17.59MeV
Energy density $= \frac{17.59 \times 10^6 \times 1.602 \times 10^{-19}}{(2.0141 + 3.016) \times 1.6605 \times 10^{-27}} = 3.37 \times 10^{14}\text{Jkg}^{-1}$
$\therefore$ tonnes to supply 10^{19}J to UK for a year = 29.6
(vi)(a) Potential energy of two protons r apart is $\frac{e^2}{4\pi\epsilon_0 r}$. If this equates to heat energy of all protons due to random, thermal motion at temperature T,

$$\begin{aligned} \frac{3}{2}k_B T &\approx \frac{e^2}{4\pi\epsilon_0 r} && (9)\\ \implies \quad T &\approx \frac{e^2}{6\pi\epsilon_0 k_B r} && (10)\\ &\approx \frac{(1.602 \times 10^{-19})^2}{6\pi \times 8.85 \times 10^{-12} \times 1.38 \times 10^{-23} \times 10^{-15}} \quad \approx 1.1 \times 10^{-10}\text{K} && (11)\end{aligned}$$

(b) The core temperature of the sun $\approx 1.5 \times 10^7$K. Now energy of photons in the sun will follow Maxwell-Boltzmann distribution $P(E) = \frac{2}{\sqrt{\pi}}(k_B T)^{-\frac{3}{2}}\sqrt{E}e^{\frac{E}{k_B T}}$ where the average energy $\approx \frac{3}{2}k_B \times 1.5 \times 10^6$J. However, an increasingly small fraction will have much higher energies, even those 1000 times higher. Since the sun is very large, even a tiny fraction may be sufficient to initiate fusion.
(vii) $B_{Pu} = 15176 \times 239 - 17.81 \times 239^{\frac{2}{3}} - \frac{0.711 \times 94^2}{239^{\frac{1}{3}}} - \frac{23.702(239 - 2 \times 94)^2}{239} = 1810.45\text{MeV}$

(viii) Pu-239 fission: ${}^{239}_{94}\text{Pu} + {}^{1}_{0}\text{n} \to {}^{134}_{54}\text{Xe} + {}^{103}_{40}\text{Zr} + 3 \times {}^{1}_{0}\text{n}$

$$\begin{aligned} B_{Zr} &= 15.76 \times 103 - 17.81 \times 103^{\frac{2}{3}} - \frac{0.711 \times 40^2}{103^{\frac{1}{3}}} - \frac{23.702(103 - 2 \times 40)^2}{103} && (12) \\ &= 867.52\text{MeV} && (13) \\ \Delta E &= B_{Xe} + B_{Zr} - B_{Pu} && (14) \\ &= 8.4137 \times 134 + 867.52 - 1810.45 && (15) \\ &= 184.51\text{MeV} && (16) \end{aligned}$$

So energy released $= \frac{184.51\times10^6\times1.602\times10^{-19}}{239\times1.6605\times10^{-27}} = 7.45 \times 10^3\text{Jkg}^{-1}$
(ix)(a) $\Delta E = 2B_{He} - B_D - B_{Li} = 2 \times 27.27 - 1.7659 - 23.3624 = 29.41\text{MeV}$, this is 31% higher than the actual value.
(b) ${}^{2}_{1}\text{H} + {}^{6}_{3}\text{Li}$ is likely to suffer much more coulomb repulsion than ${}^{2}_{1}\text{H} + {}^{3}_{1}\text{H}$ due to ${}^{6}_{3}\text{Li}$ having three times the number of protons as ${}^{3}_{1}\text{H}$. Although the shell structure of the nucleus will make the exact calculation complicated, we might expect the potential energy to be 3 times higher and the temperature for fusion to be three times higher.
(x) If one ignores the pairing term, and if $A = 2Z$, asymmetry term vanishes too, so SEMF is:

$$\begin{aligned} \frac{B}{A} &= a_v - a_s A^{-\frac{1}{3}} - a_c Z^2 A^{-\frac{4}{3}} = a_v - a_s 2^{-\frac{1}{3}} Z^{-\frac{1}{3}} - a_c 2^{-\frac{4}{3}} Z^{\frac{2}{3}} && (17) \\ \implies \frac{\mathrm{d}(\frac{B}{A})}{\mathrm{d}Z} &= \frac{1}{3} a_s 2^{-\frac{1}{2}} Z^{-\frac{4}{3}} - \frac{2}{3} a_c 2^{-\frac{4}{3}} Z^{-\frac{1}{3}} && (18) \end{aligned}$$

Maxima of $\frac{B}{A}$ is when $\frac{\mathrm{d}(\frac{B}{A})}{\mathrm{d}Z} = 0$

$$\begin{aligned} \implies \frac{1}{3} a_s 2^{-\frac{1}{2}} Z^{-\frac{4}{3}} &= \frac{2}{3} a_c 2^{-\frac{4}{3}} Z^{-\frac{1}{3}} && (19) \\ \implies Z &= \frac{a_s}{a_c} = \frac{17.81}{0.711} = 25 && (20) \end{aligned}$$

$\therefore {}^{55}_{25}\text{Mn}$ is the most stable isotope of Manganese.
2. For ${}^{240}_{94}\text{Pu}$, $B = 1817.17\text{MeV}$.
For ${}^{239}_{94}\text{Pu}$, $B = 1810.45\text{MeV}$.
If all binding energy change results in energy of γ ray, then $E_\gamma = \Delta B = 6.72\text{MeV}$, so since $E_\gamma = \frac{hc}{\gamma}$, the smaller γ is when $E_\gamma =$ all the ΔB. Using SEMF, $\gamma_{\min} = \frac{6.63\times10^{-34}\times2.998\times10^8}{6.72\times10^6\times1.602\times10^{-19}} = 1.85 \times 10^{-13}\text{m}$
4. Assume $\Delta E = 17.59\text{MeV}$ per reaction is the KE of the ${}^{4}_{2}\text{He}$ nuclei, which are ionised, relativistic calculation for speed of ${}^{4}_{2}\text{He}$ ions is

$$\begin{aligned} (1 - \frac{v^2}{c^2})^{-\frac{1}{2}} mc^2 &= \Delta E + mc^2 && (21) \\ \implies v &= c\sqrt{1 - \frac{1}{1 + \frac{\Delta E}{mc^2}}} && (22) \\ & && (23) \end{aligned}$$

So $\frac{\Delta E}{mc^2} = \frac{17.59\times10^6\times1.602\times10^{-19}}{4.00\times1.6605\times10^{-27}\times(2.998\times10^8)^2} = 4.72 \times 10^{-3}$
Therefore $v = 0.069c = 2.055 \times 10^7\text{ms}^{-1}$
If $M\text{kgs}^{-1}$ are used, then the numebr of reactions is $\frac{M}{5u}$. So thrust $F = M \times \frac{4}{5} \times 2.055 \times 10^7$, i.e. $M = \frac{43.7\times10^3}{\frac{4}{5}\times2.055\times10^7} = 2.66\text{gs}^{-1}$ is needed to supply equivalent trust for a fusion engine.